CHRISTIAN OMAR PAYÁN TORRÓNTEGUI

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SUMMARY

Robotics & Digital Systems Engineering student (8th semester) with experience in electrical schematic capture, single-line diagrams, custom PCB design, and CAD modeling. Skilled in Bill of Materials (BOM) creation, technical specification sheet evaluation, and hardware documentation.

EDUCATION

B.S. Robotics and Digital Systems Graduating Expected Dec 2026
Tecnológico de Monterrey
Zapopan, Jalisco
Relevant coursework: Electrical Circuits & Schematic Capture, Embedded Systems & PCB Design, Advanced CAD & Mechanical Design, Microsoft Office 100-Hour Certification (Excel, Data Analysis)

TECHNICAL SKILLS

CAD & Electrical Engineering: AutoCAD, SOLIDWORKS, Fusion 360, Eagle PCB, Single-Line Diagrams (Diagramas Unifilares), Schematics, Load Schedules
Documentation & Productivity: Bill of Materials (BOM) Generation, Datasheet Verification, Excel (Advanced Data Filtering/Analysis, 100-Hour Cert), Technical Documentation
Languages & Core Tools: Spanish (Native), English (B2 Professional), Python, C++, Linux, Git

PROFESSIONAL EXPERIENCE

Industrial Mounts & Support Interfaces, Zapopan, Jalisco: Hardware & Mechanical Design Engineer Feb-Jun 2026

- Engineered mechanical and electrical enclosure interfaces, performing load distribution analysis and thermal evaluation for custom mini PC deployments in industrial environments.
- Developed 3-DOF mechanical positioning mounts, generating full Bill of Materials (BOM) and reducing fabrication costs by 95% vs commercial off-the-shelf solutions.

ACADEMIC & TECHNICAL PROJECTS

DigiMix — Hardware & Schematic Integration Ago-Dic 2024
Eagle PCB, C++, Dual-Core Microcontrollers

- Designed custom schematic diagrams, unifilar electrical connections, and printed circuit board layouts for multi-sensor hardware integration.
- Managed complete component sourcing and BOM quantification; conducted technical data sheet verification for active/passive electronic components.

Port Automation & Hardware Platform Ago-Dic 2025
CAD (SOLIDWORKS/Fusion 360), Sensor Mounts

- Designed CAD models and custom hardware brackets for optical sensors, cameras, and power distribution modules on mobile robotic systems.
- Generated detailed technical documentation and assembly drawings ensuring alignment with mechanical and electrical standards.

Open-Source Medical Hardware Phantom Ago 2025 – Jun 2026
CAD/CAM, Materials Engineering, Agile

- Engineered anatomical physical model using FDM additive manufacturing (PETG/TPU), performing material cost quantification and reducing prototyping expense by 80%.